Programs on Planning and Learning

a) Implementation of Blocks World Program

Aim:

To simulate the Blocks World problem where a robot moves blocks from one configuration to another using planning.

Procedure:

1. Define the initial and goal states.
2. Apply rules for block movement.
3. Use recursive planning.

Python Code:

```
initial_state = [['A', 'B'], ['C'], []]
goal_state = [['C', 'A'], ['B'], []]

def move(state, from_stack, to_stack):
    state[to_stack].append(state[from_stack].pop())
    return state

def display_state(state):
    for i, stack in enumerate(state):
        print(f"Stack {i+1}: {stack}")
    print()

print("Initial State:")
display_state(initial_state)

# Simulate moving blocks manually (in real planning, use search)
state = move(initial_state, 1, 2)  # Move C
state = move(state, 0, 1)          # Move B
state = move(state, 0, 1)          # Move A
state = move(state, 1, 0)          # Move B
state = move(state, 2, 0)          # Move C

print("Final State:")
display_state(state)
```

Output:

Initial State:
Stack 1: ['A', 'B']
Stack 2: ['C']
Stack 3: []

Final State:
Stack 1: ['A', 'C']
Stack 2: ['B']
Stack 3: []

Result:

Thus the Blocks World program was implemented and executed successfully.